Cluster management with System Manager

ONTAP 9

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Cluster management with System Manager

Administration overview with System Manager

System Manager is an HTML5-based graphical management interface that enables you to use a web browser to manage storage systems and storage objects (such as disks, volumes, and storage tiers) and perform common management tasks related to storage systems.

The procedures in this section help you manage your cluster with System Manager in ONTAP 9.7 and later releases.

- System Manager is included with ONTAP software as a web service, enabled by default, and accessible by using a browser.
- The name of System Manager has changed beginning with ONTAP 9.6. In ONTAP 9.5 and earlier it was called OnCommand System Manager. Beginning with ONTAP 9.6 and later, it is called System Manager.
- If you are using the classic System Manager (available only in ONTAP 9.7 and earlier), refer to System Manager Classic (ONTAP 9.0 to 9.7)

Using the System Manager Dashboard, you can view at-a-glance information about important alerts and notifications, the efficiency and capacity of storage tiers and volumes, the nodes that are available in a cluster, the status of the nodes in an HA pair, the most active applications and objects, and the performance metrics of a cluster or a node.

With System Manager you can perform many common tasks, such as the following:

- Create a cluster, configure a network, and set up support details for the cluster.
- Configure and manage storage objects, such as disks, local tiers, volumes, qtrees, and quotas.
- Configure protocols, such as SMB and NFS, and provision file sharing.
- Configure protocols such as FC, FCoE, NVMe, and iSCSI for block access.
- Create and configure network components, such as subnets, broadcast domains, data and management interfaces, and interface groups.
- Set up and manage mirroring and vaulting relationships.
- Perform cluster management, storage node management, and storage virtual machine (storage VM) management operations.
- Create and configure storage VMs, manage storage objects associated with storage VMs, and manage storage VM services.
- Monitor and manage high-availability (HA) configurations in a cluster.
- Configure service processors to remotely log in, manage, monitor, and administer the node, regardless of the state of the node.

System Manager terminology

System Manager uses different terminology than the CLI for some ONTAP key functionality.

- **Local tier** – a set of physical solid-state drives or hard-disk drives you store your data on. You might know these as aggregates. In fact, if you use the ONTAP CLI, you will still see the term *aggregate* used to represent a local tier.
- **Cloud tier** – storage in the cloud used by ONTAP when you want to have some of your data off premises for one of several reasons. If you are thinking of the cloud part of a FabricPool, you've already figured it out. And if you are using a StorageGRID system, your cloud might not be off premises at all. (A cloud-like experience on premises is called a *private cloud*.)
- **Storage VM** – a virtual machine running within ONTAP that provides storage and data services to your clients. You might know this as an *SVM* or a *vserver*.
- **Network interface** - an address and properties assigned to a physical network port. You might know this as a *logical interface (LIF)*.
- **Pause** - an action that halts operations. Before ONTAP 9.8, you might have referred to *quiesce* in other versions of System Manager.

Use System Manager to access a cluster

If you prefer to use a graphic interface instead of the command-line interface (CLI) for accessing and managing a cluster, you can do so by using System Manager, which is included with ONTAP as a web service, is enabled by default, and is accessible by using a browser.

> Beginning with ONTAP 9.12.1, System Manager is fully integrated with BlueXP.
>
> With BlueXP, you can manage your hybrid multicloud infrastructure from a single control plane while retaining the familiar System Manager dashboard.
>
> See System Manager integration with BlueXP.

About this task

You can use a cluster management network interface (LIF) or node management network interface (LIF) to access System Manager. For uninterrupted access to System Manager, you should use a cluster management network interface (LIF).

Before you begin

- You must have a cluster user account that is configured with the "admin" role and the "http" and "console" application types.
- You must have enabled cookies and site data in the browser.

Steps

1. Point the web browser to the IP address of the cluster management network interface:
   - If you are using IPv4: **`https://`*`cluster-mgmt-LIF`***
   - If you are using IPv6: **`https://[`*`cluster-mgmt-LIF`*`]`**

     > Only HTTPS is supported for browser access of System Manager.

   If the cluster uses a self-signed digital certificate, the browser might display a warning indicating that the certificate is not trusted. You can either acknowledge the risk to continue the access or install a Certificate

Authority (CA) signed digital certificate on the cluster for server authentication.

2. **Optional:** If you have configured an access banner by using the CLI, then read the message that is displayed in the **Warning** dialog box, and choose the required option to proceed.

   This option is not supported on systems on which Security Assertion Markup Language (SAML) authentication is enabled.

   - If you do not want to continue, click **Cancel**, and close the browser.
   - If you want to continue, click **OK** to navigate to the System Manager login page.
3. Log in to System Manager by using your cluster administrator credentials.

   > Beginning with ONTAP 9.11.1, when you log in to System Manager, you can specify the locale. The locale specifies certain localization settings, such as language, currency, time and date format, and similar settings. For ONTAP 9.10.1 and earlier, the locale for System Manager is detected from the browser. To change the locale for System Manager, you have to change the locale of the browser.

4. **Optional**: Beginning with ONTAP 9.12.1, you can specify your preference for the appearance of System Manager:
   a. In the upper right corner of System Manager, click to manage user options.
   b. Position the **System Theme** toggle switch to your preference:

Toggle position	Appearance setting
(left)	Light theme (Light background with dark text)
OS (center)	Default to the theme preference that was set for the operating system's applications (usually the theme setting for the browser that is used to access System Manager).
(right)	Dark theme (Dark background with light text)

Related information

Managing access to web services

Accessing a node's log, core dump, and MIB files by using a web browser

Enable new features by adding license keys

In releases earlier than ONTAP 9.10.1, ONTAP features are enabled with license keys, and features in ONTAP 9.10.1 and later are enabled with a NetApp license file. You can add license keys and NetApp license files using System Manager.

Beginning with ONTAP 9.10.1, you use System Manager to install a NetApp License File to enable multiple licensed features all at once. Using a NetApp License File simplifies license installation because you no longer have to add separate feature license keys. You download the NetApp License File from the NetApp Support Site.

If you already have license keys for some features and you are upgrading to ONTAP 9.10.1, you can continue to use those license keys.

Steps

1. Select **Cluster > Settings**.
2. Under **Licenses**, select →.
3. Select **Browse**. Choose the NetApp License File you downloaded.
4. If you have license keys you want to add, select **Use 28-character license keys** and enter the keys.

Download a cluster configuration

Beginning with ONTAP 9.11.1, you can use System Manager to download the configuration of a cluster.

Steps

1. Click **Cluster > Overview**.
2. Click ⋮ More to display the drop-down menu.
3. Select **Download configuration**.
4. Select the HA pairs, then click **Download**.

   The configuration is downloaded as an Excel spreadsheet.

   - The first sheet contains cluster details.
   - The other sheets contain node details.

Assign tags to a cluster

Beginning with ONTAP 9.14.1, you can use System Manager to assign tags to a cluster to identify objects as belonging to a category, such as projects or cost centers.

About this task

You can assign a tag to a cluster. First, you need to define and add the tag. Then, you can also edit or delete the tag.

Tags can be added when you create a cluster, or they can be added later.

You define a tag by specifying a key and associating a value to it using the format “key:value”. For example: “dept:engineering” or “location:san-jose”.

The following should be considered when you create tags:

- Keys have a minimum length of one character and cannot be null. Values can be null.
- A key can be paired with multiple values by separating the values with a comma, for example, “location:san-jose,toronto”
- Tags can be used for multiple resources.
- Keys must start with a lowercase letter.

Steps

To manage tags, performing the following steps:

1. In System Manager, click **Cluster** to view the overview page.

   The tags are listed in the **Tags** section.

2. Click **Manage Tags** to modify existing tags or add new ones.

   You can add, edit, or delete the tags.

To perform this action…	Perform these steps…
Add a tag	a. Click **Add Tag**. b. Specify a key and its value or values (separate multiple values with commas). c. Click **Save**.
Edit a tag	a. Modify the content in the **Key** and **Values (optional)** fields. b. Click **Save**.
Delete a tag	a. Click 🗑 next to the tag you want to delete.

View and submit support cases

Beginning with ONTAP 9.9.1, you can view support cases from Active IQ associated with the cluster. You can also copy cluster details that you need to submit a new support case on the NetApp Support Site.
Beginning with ONTAP 9.10.1, you can enable telemetry logging, which helps support personnel troubleshoot problems.

> To receive alerts about firmware updates, you must be registered with Active IQ Unified Manager. Refer to Active IQ Unified Manager documentation resources.

Steps

1. In System Manager, select **Support**.

   A list of open support cases associated with this cluster is displayed.

2. Click on the following links to perform procedures:
   - **Case Number**: See details about the case.
   - **Go to NetApp Support Site**: Navigate to the **My AutoSupport** page on the NetApp Support Site to view knowledge base articles or submit a new support case.
   - **View My Cases**: Navigate to the **My Cases** page on the NetApp Support Site.
   - **View Cluster Details**: View and copy information you will need when you submit a new case.

Enable telemetry logging

Beginning with ONTAP 9.10.1, you can use System Manager to enable telemetry logging. When telemetry logging is allowed, messages that are logged by System Manager are given a specific telemetry identifier that indicates the exact process that triggered the message. All messages that are issued relating to that process have the same identifier, which consists of the name of the operational workflow and a number (for example "add-volume-1941290").

If you experience performance problems, you can enable telemetry logging, which allows support personnel to more easily identify the specific process for which a message was issued. When telemetry identifiers are added to the messages, the log file is only slightly enlarged.

Steps

1. In System Manager, select **Cluster > Settings**.
2. In **UI Settings** section, click the check box for **Allow telemetry logging**.

Manage the maximum capacity limit of a storage VM in System Manager

Beginning with ONTAP 9.13.1, you can use System Manager to enable a maximum capacity limit for a storage VM and set a threshold to trigger alerts when the used storage reaches a certain percentage of the maximum capacity.

Enable a maximum capacity limit for a storage VM

Beginning with ONTAP 9.13.1, you can specify the maximum capacity that can be allocated for all volumes in a storage VM. You can enable the maximum capacity when you add a storage VM or when you edit an existing storage VM.

Steps

1. Select **Storage** > **Storage VMs**.
2. Perform one of the following:
   - To add a storage VM, click + Add .
   - To edit a storage VM, click ⋮ next to the name of the storage VM, and then click **Edit**.
3. Enter or modify the settings for the storage VM, and select the check box labeled "Enable maximum capacity limit".
4. Specify the maximum capacity size.
5. Specify the percentage of the maximum capacity you want to use as a threshold to trigger alerts.
6. Click **Save**.

Edit the maximum capacity limit of a storage VM

Beginning with ONTAP 9.13.1, you can edit the maximum capacity limit of an existing storage VM, if the maximum capacity limit has been enabled already.

Steps

1. Select **Storage** > **Storage VMs**.
2. Click ⋮ next to the name of the storage VM, and then click **Edit**.

   The check box labeled "Enable maximum capacity limit" is already checked.

3. Perform one of the following steps:

Action	Steps
Disable the maximum capacity limit	1. Uncheck the check box. 2. Click **Save**.
Modify the maximum capacity limit	1. Specify the new maximum capacity size. (You cannot specify a size that is less than the already allocated space in the storage VM.) 2. Specify the new percentage of the maximum capacity you want to use as a threshold to trigger alerts. 3. Click **Save**.

Related information

- View the maximum capacity limit of a storage VM
- Capacity measurements in System Manager
- Manage SVM capacity limits using the ONTAP CLI

Monitor capacity in System Manager

Using System Manager, you can monitor how much storage capacity has been used and how much is still available for a cluster, a local tier, or a storage VM.

With each version of ONTAP, System Manager provides more robust capacity monitoring information:

- Beginning with ONTAP 9.10.1, System Manager lets you view historical data about the cluster's capacity and projections about how much capacity will be used or available in the future. You can also monitor the capacity of local tiers and volumes.
- Beginning with ONTAP 9.12.1, System Manager displays the amount of committed capacity for a local tier.
- Beginning with ONTAP 9.13.1, you can enable a maximum capacity limit for a storage VM and set a threshold to trigger alerts when the used storage reaches a certain percentage of the maximum capacity.

> Measurements of used capacity are displayed differently depending on your ONTAP version. Learn more in Capacity measurements in System Manager.

View the capacity of a cluster

You can view capacity measurements for a cluster on the Dashboard in System Manager.

Before you begin

To view data related to the capacity in the cloud, you must have an account with Active IQ Digital Advisor and

be connected.

Steps

1. In System Manager, click **Dashboard**.
2. In the **Capacity** section, you can view the following:
   - Total used capacity of the cluster
   - Total available capacity of the cluster
   - Percentages of used and available capacity.
   - Ratio of data reduction.
   - Amount of capacity used in the cloud.
   - History of capacity usage.
   - Projection of capacity usage

   > In System Manager, capacity representations do not account for root storage tier (aggregate) capacities.

3. Click the chart to view more details about the capacity of the cluster.

   Capacity measurements are shown in two bar charts:

   - The top chart displays the physical capacity: the size of physical used, reserved, and available space.
   - The bottom chart displays the logical capacity: the size of client data, Snapshot copies, and clones, and the total logical used space.

   Below the bar charts are measurements for data reduction:

   - Data reduction ratio for only the client data (Snapshot copies and clones are not included).
   - Overall data reduction ratio.

For more information, see Capacity measurements in System Manager.

View the capacity of a local tier

You can view details about the capacity of local tiers. Beginning with ONTAP 9.12.1, the **Capacity** view also includes the amount of committed capacity for a local tier, enabling you to determine whether you need to add capacity to the local tier to accommodate the committed capacity and avoid running out of free space.

Steps

1. Click **Storage > Tiers**.
2. Select the name of the local tier.
3. On the **Overview** page, in the **Capacity** section, the capacity is show in a bar chart with three measurements:
   - Used and reserved capacity
   - Available capacity
   - Committed capacity (beginning with ONTAP 9.12.1)

4. Click the chart to view details about the capacity of the local tier.

   Capacity measurements are shown in two bar charts:

   - The top bar chart displays physical capacity: the size of physical used, reserved, and available space.
   - The bottom bar chart displays logical capacity: the size of client data, Snapshot copies, and clones, and the total of logical used space.

   Below the bar charts are measurements ratios for data reduction:

   - Data reduction ratio for only the client data (Snapshot copies and clones are not included).
   - Overall data reduction ratio.

For more information, see Capacity measurements in System Manager.

Optional actions

- If the committed capacity is larger than the capacity of the local tier, you might consider adding capacity to the local tier before it runs out of free space. See Add capacity to a local tier (add disks to an aggregate).
- You can also view the storage that specific volumes use in the local tier by selecting the **Volumes** tab.

View the capacity of the volumes in a storage VM

You can view how much storage is used by the volumes in a storage VM and how much capacity is still available. The total measurement of used and available storage is called "capacity across volumes".

Steps

1. Select **Storage** > **Storage VMs**.
2. Click on the name of the storage VM.
3. Scroll to the **Capacity** section, which shows a bar chart with the following measurements:

   - **Physical used**: Sum of physical used storage across all volumes in this storage VM.
   - **Available**: Sum of available capacity across all volumes in this storage VM.
   - **Logical used**: Sum of logical used storage across all volumes in this storage VM.

For more details about the measurements, see Capacity measurements in System Manager.

View the maximum capacity limit of a storage VM

Beginning with ONTAP 9.13.1, you can view the maximum capacity limit of a storage VM.

Before you begin

You must enable the maximum capacity limit of a storage VM before you can view it.

Steps

1. Select **Storage** > **Storage VMs**.

   You can view the maximum capacity measurements in two ways:

   - In the row for the storage VM, view the **Maximum Capacity** column which contains a bar chart that shows the used capacity, available capacity, and maximum capacity.

- Click the name of the storage VM. On the **Overview** tab, scroll to view the maximum capacity, allocated capacity, and capacity alert threshold values in the left column.

Related information

- Edit the maximum capacity limit of a storage VM
- Capacity measurements in System Manager

View hardware configurations to determine problems

Beginning with ONTAP 9.8, you can use System Manager to view the configuration of hardware on your network and determine the health of your hardware systems and cabling configurations.

Steps

To view hardware configurations, perform the following steps:

1. In System Manager, select **Cluster > Hardware**.
2. Hover your mouse over components to view status and other details.

   You can view various types of information:

   - Information about controllers
   - Information about disk shelves
   - Information about storage switches
3. Beginning with ONTAP 9.12.1, you can view cabling information in System Manager. Click the **Show Cables** check box to view cabling, then hover over a cable to view its connectivity information.
   - Information about cabling

Information about controllers

You can view the following:

Nodes

Nodes:

- You can view the front and rear views.
- For models with an internal disk shelf, you can also view the disk layout in the front view.
- You can view the following platforms:

Platform	Supported in System Manager in ONTAP version…						
	9.14.1	9.13.1	9.12.1	9.11.1	9.10.1	9.9.1	9.8 (preview mode only)
AFF A150	Yes	Yes					
AFF A220	Yes	Yes	Yes	Yes	Yes	Yes	Yes
AFF A250	Yes	Yes	Yes	Yes	Yes	Yes	
AFF A300	Yes	Yes	Yes	Yes	Yes	Yes	Yes
AFF A320	Yes	Yes	Yes	Yes	Yes	Yes	
AFF A400	Yes	Yes	Yes	Yes	Yes	Yes	Yes
AFF A700	Yes	Yes	Yes	Yes	Yes	Yes	Yes
AFF A700s	Yes	Yes	Yes	Yes	Yes	Yes	
AFF A800	Yes	Yes	Yes	Yes	Yes	Yes	
AFF C190	Yes	Yes	Yes	Yes	Yes	Yes	Yes
AFF C250	Yes	Yes	Yes *	Yes *	Yes *		
AFF C400	Yes	Yes	Yes *	Yes *	Yes *		
AFF C800	Yes	Yes	Yes *	Yes *	Yes *		
ASA A150	Yes	Yes					
ASA A250	Yes	Yes					
ASA A400	Yes	Yes					

ASA A800	Yes	Yes					
ASA A900	Yes	Yes					
ASA C250	Yes	Yes					
ASA C400	Yes	Yes					
ASA C800	Yes	Yes					
FAS500f	Yes	Yes	Yes	Yes	Yes	Yes	
FAS2720	Yes	Yes	Yes	Yes			
FAS2750	Yes	Yes	Yes	Yes			
FAS8300	Yes	Yes	Yes	Yes			
FAS8700	Yes	Yes	Yes	Yes			
FAS9000	Yes	Yes	Yes	Yes			
FAS9500	Yes	Yes	Yes	Yes			
* Install the latest patch releases to view these devices.							

Ports

Ports:

- You will see a port highlighted in red if it is down.
- When you hover over the port, you can view the status of a port and other details.
- You cannot view console ports.

  Notes:

  - For ONTAP 9.10.1 and earlier, you will see SAS ports highlighted in red when they are disabled.
  - Beginning with ONTAP 9.11.1, you will see SAS ports highlighted in red only if they are in an error state or if a cabled port that is being used goes offline. The ports appear in white if they are offline and uncabled.

FRUs

FRUs:

Information about FRUs appears only when the state of a FRU is non-optimal.

- Failed PSUs in nodes or chassis.

- High temperatures detected in nodes.
- Failed fans on the nodes or chassis.

Adapter cards

Adapter cards:

- Cards with defined part number fields display in the slots if external cards have been inserted.
- Ports display on the cards.
- For a supported card, you can view images of that card. If the card is not in the list of supported part numbers, then a generic graphic appears.

Information about disk shelves

You can view the following:

Disk shelves

Disk shelves:

- You can display the front and rear views.
- You can view the following disk shelf models:

If your system is running…	Then you can use System Manager to view…
ONTAP 9.9.1 and later	All shelves that have *not* been designated as "end of service" or "end of availability"
ONTAP 9.8	DS4243, DS4486, DS212C, DS2246, DS224C, and NS224

Shelf ports

Shelf ports:

- You can view port status.
- You can view remote port information if the port is connected.

Shelf FRUs

Shelf FRUs:

- PSU failure information displays.

Information about storage switches

You can view the following:

Storage switches

Storage switches:

- The display shows switches that act as storage switches used to connect shelves to nodes.
- Beginning with ONTAP 9.9.1, System Manager displays information about a switch that acts as both a storage switch and a cluster, which can also be shared between nodes of an HA pair.
- The following information displays:
  - Switch name
  - IP address
  - Serial number
  - SNMP version
  - System version
- You can view the following storage switch models:

If your system is running…	Then you can use System Manager to view…
ONTAP 9.11.1 or later	Cisco Nexus 3232C Cisco Nexus 9336C-FX2 Mellanox SN2100
ONTAP 9.9.1 and 9.10.1	Cisco Nexus 3232C Cisco Nexus 9336C-FX2
ONTAP 9.8	Cisco Nexus 3232C

Storage switch ports

Storage switch ports

- The following information displays:
  - Identity name
  - Identity index
  - State
  - Remote connection
  - Other details

Information about cabling

Beginning with ONTAP 9.12.1, you can view the following cabling information:

- **Cabling** between controllers, switches, and shelves when no storage bridges are used
- **Connectivity** that shows the IDs and MAC addresses of the ports on either end of the cable

Manage nodes using System Manager

Using System Manager, you can add nodes to a cluster and rename them. You can also reboot, take over, and give back nodes.

Add nodes to a cluster

You can increase the size and capabilities of your cluster by adding new nodes.

Before you Start

You should have already cabled the new nodes to the cluster.

About this task

There are separate processes for working with System Manager in ONTAP 9.7 or ONTAP 9.8 and later.

ONTAP 9.8 and later procedure

Adding nodes to a cluster with System Manager (ONTAP 9.8 and later)

Steps

1. Select **Cluster > Overview**.

   The new controllers are shown as nodes connected to the cluster network but are not in the cluster.

2. Select **Add**.
   - The nodes are added into the cluster.
   - Storage is allocated implicitly.

ONTAP 9.7 procedure

Adding nodes to a cluster with System Manager (ONTAP 9.7)

Steps

1. Select **(Return to classic version)**.
2. Select **Configurations > Cluster Expansion**.

   System Manager automatically discovers the new nodes.

3. Select **Switch to the new experience**.
4. Select **Cluster > Overview** to view the new nodes.

Shut down, reboot or edit service processor

When you reboot or shutdown a node, its HA partner automatically executes a takeover.

Steps

1. Select **Cluster > Overview**.
2. Under **Nodes**, select ⋮.
3. Select the node and then select **Shut down**, **Reboot**, or **Edit Service Processor**.

If a node has been rebooted and is waiting for giveback, the **Giveback** option is also available.

If you select **Edit Service Processor**, you can choose **Manual** to input the IP address, subnet mask and gateway, or you can choose **DHCP** for dynamic host configuration.

Rename nodes

Beginning with ONTAP 9.14.1, you can rename a node from the cluster overview page.

Steps

1. Select **Cluster**. The cluster overview page displays.
2. Scroll down to the **Nodes** section.
3. Next to the node that you want to rename, select ⋮, and select **Rename**.
4. Modify the node name, and then select **Rename**.

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